

Neuroimaging Evidence of Motor Control and Pain Processing in the Human Midcingulate Cortex

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Human neuroimaging and virus-tracing studies in monkey predict that motor control and pain processes should overlap in anterior midcingulate cortex (aMCC), but there is currently no direct evidence that this is the case. We used a novel functional magnetic resonance imaging paradigm to examine brain activity while subjects performed a motor control task, experienced a pain-eliciting stimulus on their hand, and performed the motor control task while also experiencing the pain-eliciting stimulus. Our experiment produced 3 novel results. First, group-level analyses showed that when separate trials of motor control and pain processing were performed, overlapping functional activity was found in the same regions of aMCC, supplementary motor area (SMA), anterior insula, and putamen. Secondly, increased activity was found in the aMCC and SMA when motor control and pain processing occurred simultaneously. Thirdly, individual-level analyses showed that 93% of subjects engaged the same region of aMCC during separate trials of motor control and pain processing irrespective of differences in the sulcal/gyral morphology of the cingulate cortex across individuals. These observations provide direct evidence in humans that the same region of aMCC is engaged for motor control and pain processing.

Keywords: cingulate cortex, fMRI, motor, pain, supplementary motor area

Introduction

Movements influence the way we experience pain, and pain alters the way that we control movement. Identifying the candidate brain areas that underlie this relationship is important for the development of novel targeted interventions in rehabilitation and performance contexts. It is well established from separate motor control and pain processing studies that anterior midcingulate cortex (aMCC) is engaged during the planning and execution of motor function (Luppino et al. 1991; Picard and Strick 2001; Coxon et al. 2010; Amiez and Petrides 2014; Hoffstaedter et al. 2013), and when experiencing pain-eliciting stimuli (Vogt et al. 1996; Apkarian et al. 2000; Wager et al. 2004; Lui et al. 2008; Duerden and Albanese 2013; Wager et al. 2013). Although virus-tracing studies in monkey predict that motor control and pain processes should overlap in the aMCC (Dum et al. 2009), there is currently no direct evidence that this is the case. The goals in the current study are to determine whether the same regions of aMCC are engaged during motor control and pain processing in humans, whether this overlap in function is robust at the individual level, and whether it is influenced by differences in anatomical morphology of the cingulate cortex across individuals.

The MCC is divided into anterior (aMCC) and posterior (pMCC) regions by a coronal plane positioned at the anterior commissure. Based on physiologic and anatomic criteria in

monkey, the rostral cingulate zone (RCZ) lies in the aMCC and is considered a somatotopically organized premotor area that is involved in motor planning and inhibition (Picard and Strick 1996; Dum and Strick 2005; Morecraft and Tanji 2009). The caudal cingulate zone is positioned at the boundary of aMCC and pMCC and is involved in orienting the body in response to sensory stimuli and motor execution (Vogt 2005). Over the last 2 decades, the functional segregation model of cingulate cortex has associated anterior cingulate cortex (ACC) with affective processing, aMCC with cognitive control and fear avoidance, and pMCC with skeletomotor function (Bush et al. 2000; Vogt 2005). However, several recent studies have associated aMCC with motor function (Dum et al. 2009; Mutschler et al. 2009), pain processing (Vogt 2005; Peltz et al. 2011), cognitive control (Nee et al. 2011), and emotion processing (Kober et al. 2008), leading some to question the segregation model (Shackman et al. 2011).

In the context of the relationship between motor control and pain, evidence for integrated function in the aMCC comes from virus-tracing evidence in monkey that identified RCZ as a region that receives projections from the spinothalamic system, which is the major pathway for transmitting nociceptive information to the cortex (Dum et al. 2009). The same authors presented a meta-analysis of human brain imaging studies that separately manipulated either motor control or pain processing. A predominant overlap was identified on the medial wall of the hemisphere at the boundary of aMCC and pMCC. However, this finding is inconsistent with previous evidence, which showed that motor control and pain processes are segregated in the aMCC (Kwan et al. 2000). In addition, evidence from the meta-analysis is not direct because it reflects the amalgamation of group data, and could be influenced by different experimental paradigms (e.g., location and duration of pain-eliciting stimulus; type of motor task), and by between study differences in neuroimaging analyses, such as spatial filtering and normalization procedures. To determine whether the same region of aMCC is active during motor control and pain processing, the most direct approach is to examine brain activity in the same individual while controlling for the limb performing the motor task and the limb to which the pain-eliciting stimulus is delivered. It is also important to match the duration of each task and assess both motor performance and pain ratings. Finally, previous evidence shows that functional activity in the aMCC is somatotopically organized, and that this organization is influenced by differences in the sulcal/gyral morphology of the cingulate cortex (Amiez and Petrides 2014). Hence, analyses must be conducted at the individual level when taking cingulate cortex morphology into account. In the current study, we address each of these issues.

Based on evidence from separate studies of motor control and pain processing, we hypothesized that functional activity would be evident in the same region of aMCC at the group level when the same individual produces force or experiences pain in separate trials. We also hypothesized that increased functional activity would be evident in the aMCC when motor control and pain processing are coupled in the same trial. Finally, we expected functional activity in the same region of aMCC during motor control and pain processing to be robust at the individual-subject level, even when taking into account differences in the sulcal/gyral morphology of the cingulate cortex across individuals.

Methods

Subjects

Fifteen healthy right-handed adult subjects with normal or corrected-to-normal vision were recruited (8 females, 7 males; $M = 27.4$ years, range: 19–56 years). Each subject provided informed consent to all procedures, which were approved by the local Institutional Review Board and were in accordance with the Declaration of Helsinki. All subjects were pre-screened for contraindications to magnetic resonance imaging (MRI), such as pregnancy, claustrophobia, and metallic implants, as well as any history of neurologic disease, chronic pain, or current acute pain. Since depression and anxiety have been shown to influence how experimental pain is experienced (Rhudy and Meagher 2000; Weiss et al. 2011), all subjects completed the State and Trait segments of the STAI Anxiety Inventory (STAI-S, STAI-T: Spielberger 1983), the Beck Depression Inventory (BDI: Beck and Steer 1987), Pain Catastrophizing Scale (PCS: McCracken et al. 1992), and the Tampa Scale of Kinesiophobia (TSK: Kori et al. 1990). All scores were within the normal range of responses given by adult subjects (STAI-S: $M = 27.4$, range = 20–41; STAI-T: $M = 29.8$, range = 21–42; BDI: $M = 1.69$, range = 0–6; PCS: $M = 8.56$, range = 0–22; TSK: $M = 30.06$, range = 23–39).

Force Data Acquisition

During the practice session, each subject's maximum voluntary contraction (MVC) was measured using a pinch-grip-force transducer (Jamar Hydraulic Pinch Gauge). Subjects were asked to use their right hand to produce the maximum force contraction for 3 consecutive 5-s trials. The trials were separated by a 60-s period of rest. The average of the 3 peak force levels was accepted as the MVC for each subject. Subjects were required to produce 15% of their MVC during the experiment. Setting a target force level based on an individual's MVC normalizes the amplitude of force between subjects which is important because force amplitude can influence the blood oxygen level-dependent (BOLD) signal in regions, such as M1, thalamus, and cerebellum (Spraker et al. 2007; 2012). Inside the magnet, force was produced by the right hand against a custom-built, MRI-compatible, fiber-optic transducer, which uses Fiber Bragg Grating technology and has a resolution of 0.025 N. The signal from the force transducer was transmitted to an sm130 amplifier (Micron Optics, Atlanta, GA, USA) housed in the control room. A custom-built LabVIEW software sampled the force signal at 125 Hz and displayed it on a magnetically shielded display, which was visible to the subject via a mirror installed in the head coil, providing the subject with real-time feedback of their force performance. The display that subjects viewed while in the scanner is shown in Figure 1A. The timing of the task was synchronized with the scanner acquisition by triggering the LabVIEW program with a Transistor-Transistor Logic (TTL) pulse generated by the scanner at the start of each scan.

Thermal Stimulation

Thermal stimuli were delivered to the thenar eminence of the right hand using a 573-mm² CHEPS thermode (PATHWAY System, Medoc Advanced Medical Systems, Chapel Hill, NC, USA). The CHEPS thermode heats up at a rate of 70 °C/s and cools down at a rate of 40 °C/s.

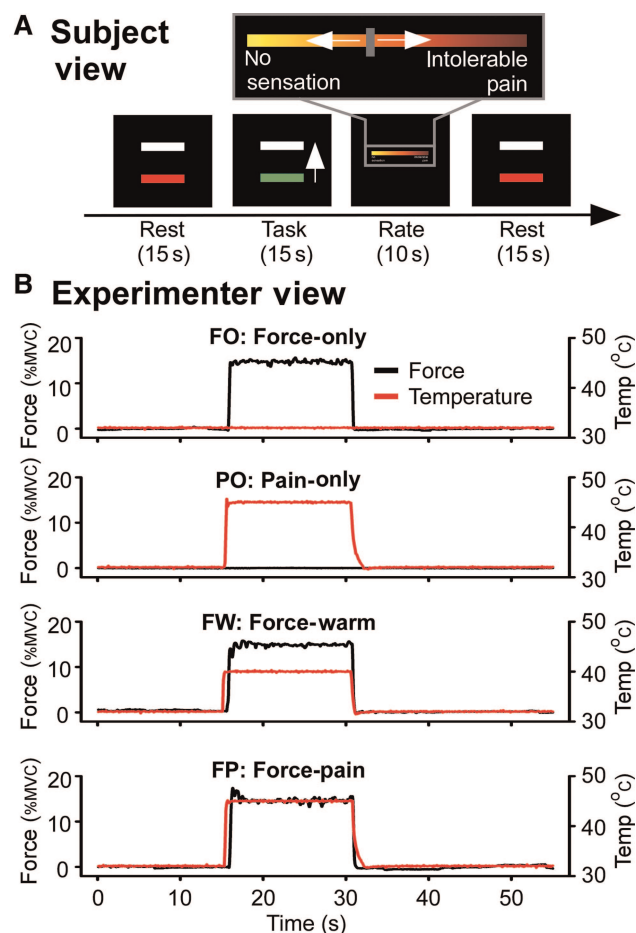


Figure 1. Experimental paradigm. (A) The red bar indicated a rest period. When the bar turned green, subjects produced a sustained isometric contraction for 15 s in the FO, FW, and FP conditions. The green bar moved in response to the force produced, thus providing real-time visual feedback. In the PO condition, subjects did not produce force, but the bar turned green and moved based on a prerecorded force-trace from the same subject. Therefore, visual information was similar across all conditions. Subjects completed 4 separate functional scans. Each scan had 6 trials of the same condition (FO, FW, FP or PO). Subjects were informed about the condition they would complete before each scan. After each trial, subjects had 10 s to rate the level of sensation they felt during the preceding 15-s trial. (B) Examples of the force-trace produced by one subject overlaid on the temperature-trace during each trial type. The black lines (left y-axis) denote the force output, and the red lines (right y-axis) denote the temperature of the thermode for each experimental condition.

The baseline temperature was set at 32 °C (Moulton et al. 2011). Hence, the thermode reached the requisite pain-eliciting temperatures in about 0.2 s.

Thermal Calibration

The temperatures needed to elicit warm and moderately painful sensations were determined for each subject using a calibration paradigm that each subject completed during the practice session. To keep the calibration session consistent with the experimental trials, subjects were exposed to 15 s of stimulation separated by 30 s of rest. After each stimulation period, subjects made a verbal rating on a visual scale anchored with numbers from 0 to 10 accompanied by the descriptors “no sensation” and “intolerable pain.” Subjects were informed that they could give fractional ratings. Following each stimulation period, subjects were asked to rate the stimulation and whether they could tolerate a stronger stimulation. The target temperature of the thermode was raised by 1–2 °C in successive 15-s blocks depending on the comfort level of the subject. The temperature corresponding to a rating of 2–3

(warm but no pain) was used in the warm trials, and that corresponding to a rating of 5–7 (moderate pain) was used in the pain trials.

Experimental Paradigm

The experiment was designed to examine the unique and overlapping brain areas that underlie force production and pain processing (Fig. 1). Subjects (1) produced visually guided grip-force with their right hand without any thermal stimulation (force-only: FO), (2) experienced pain-eliciting thermal stimulation on the thenar eminence of their right hand without producing force (pain-only: PO), (3) produced visually guided grip-force with their right hand while simultaneously experiencing warm thermal stimulation on the thenar eminence of the same hand (force-warm: FW), and (4) produced visually guided grip-force with their right hand while simultaneously experiencing pain-eliciting thermal stimulation on the thenar eminence of the same hand (force-pain: FP). It has previously been shown that uncertainty about the timing and magnitude of thermal stimulation can change subjective and physiologic responses to pain (Oka et al. 2010). To eliminate such uncertainties, subjects were instructed that the same temperature would be used during PO and FP trials, and that the onset of the thermal stimuli would be the same across conditions. The FW condition was included to control for the simultaneity of 2 processes in the FP condition: force production and thermal stimulation. Subjects completed one scan of each experimental condition. Each scan included 6 blocks of the same condition. Experimental conditions were counter-balanced across subjects. During rest, subjects did not produce force or experience stimulation and fixated on the white target bar and the red force bar while the thermode maintained a baseline temperature of 32 °C. Each 15-s block began when the force bar turned green and moved in response to the force produced by the subject. During force trials (FO, FW, and FP), subjects produced force to match the position of the green bar with the white target bar as accurately as they could. The target bar was set at 15% of each subject's MVC. Subjects were instructed not to produce force during the PO condition, and force data recorded from the transducer during the PO condition confirmed that all subjects followed this instruction. To keep the visual feedback in the PO condition consistent with other conditions, motion of the green bar recorded during the practice session was displayed on the screen during PO trials. The timing of stimulation was synchronized with the force task by triggering the PATHWAY System with a TTL pulse generated by the LabVIEW program at the start of each block when the red bar turned green. After each 15-s block, a visual analog scale (VAS), as shown in Figure 1A, was presented on the screen for subjects to rate the level of stimulation they had just experienced. The digital resolution of the VAS was 0.1. Subjects moved the slider on the VAS by pressing buttons on a button-box with their left hand. The rating period lasted 10 s and the final position of the cursor on the VAS was recorded. The rating period was followed by a 15-s rest interval.

Analysis 1: Force Data and Thermal Ratings

Force data were analyzed using custom programs in LabVIEW. Each subject's force-traces were digitally filtered through a fourth-order Butterworth filter with a 20-Hz low-pass cutoff. Motor performance was characterized by mean force amplitude, force variability (standard deviation), force error (root mean square error), and duration of force. These performance characteristics were calculated from a 12-s period within each contraction, consistent with previous experiments that have examined visually guided isometric force control (Coombes et al. 2010, 2011; Lodha et al. 2010, 2012; Neely et al. 2013). The onset of each contraction was identified as the time point where force rose above twice the baseline value. The offset of each contraction was identified as the next time point where force fell below twice the baseline value. The duration of each force contraction was calculated from successive onset and offset points. Baseline values were calculated as the mean force amplitude during the 0.3-s period prior to each block (Coombes et al. 2010, 2011). Each performance characteristic was averaged across the 6 blocks in each force condition (FO, FW, and FP) to obtain average force-dependent variables for each subject. At the group level, separate 1-way repeated-measures analysis of variance (ANOVA) were run for each force-dependent variable. Significant main

effects were followed up with *t*-tests, which were corrected for multiple comparisons using the Bonferroni correction.

Mean thermal rating scores were calculated from the 6 blocks in each condition. A 1-way repeated-measures ANOVA was then used to examine differences in ratings across the 3 stimulation conditions (PO, FW, and FP). A significant main effect of condition was followed up with *t*-tests, which were corrected for multiple comparisons using the Bonferroni correction.

Image Acquisition

MRI data were collected using a 32-channel head coil on a 3-T MRI scanner (Achieva, Philips Medical Systems, Best, The Netherlands) at the McKnight Brain Institute at the University of Florida. Anatomical images were acquired in 170 contiguous axial slices at 1 × 1 × 1 mm resolution with a T_1 -weighted fast spoiled gradient-recalled sequence (repetition time [TR] = 6.8 ms; echo time [TE] = 3.3 ms; flip angle = 8°). Functional data were acquired in 46 axial slices at 3 × 3 × 3 mm resolution using a single-shot gradient echo-planar imaging pulse sequence (TR = 2500 ms; TE = 30 ms; flip angle = 80°). Each functional scan was 265 s long. Subjects wore ear plugs during the sessions to minimize discomfort due to scanner noise. Small cushions were placed in the head coil around subject's head to minimize head-motion.

Imaging Data Analysis

Data preprocessing and analysis were performed with Analysis of Functional NeuroImages (AFNI) software (National Institute of Health, Bethesda, MD, USA) and custom UNIX shell scripts. The anatomical image of each participant was skull-stripped and warped into the standard Montreal Neurological Institute (MNI) space. The first 4 volumes of each functional scan series were discarded to allow for T_1 -equilibration effects, and the remaining volumes underwent slice acquisition-dependent slice-time correction. This was followed by registration of the functional scan volumes to a base volume (via rigid body rotations), alignment with the anatomical scan and warping into the MNI space in a single transformation to avoid repeated resampling of the data. Next, each volume was smoothed to a final smoothness of 6-mm full-width at half-maximum using AFNI's 3dBlurToFWHM program (inherent smoothness being 4.5–5 mm). To normalize the data, the signal in each voxel at each time point was scaled by the mean of its time series.

Analysis 2: Activation During Force-Only and Pain-Only Trials

The BOLD signals during the task (FO, PO, FP, and FW) and during the rating period were modeled separately by boxcar regressors convolved with the canonical hemodynamic response function. The 6 head-motion parameters (3 rotations and 3 translations) calculated during registration to a base volume were included in the general linear model as regressors-of-no-interest. When head-motion between adjacent volumes was >0.6 mm, both volumes were excluded from the regression analysis. A minimum of 34 volumes per subject per condition (94%) remained after excluding motion-affected volumes.

The resulting regression-coefficient (beta-value) maps for each condition were averaged across subjects, and *t*-statistics were calculated for each voxel to generate group-level statistical maps for each condition. Family-wise error rate (FWER) in the statistical maps was maintained <0.05 by rejecting voxels with *P*-values >0.005 and clusters <567 mm³ in volume (Forman et al. 1995). The *P*-value threshold and the extent threshold required to achieve the reported FWER were determined using AFNI's 3dClustSim program, which takes the average smoothness of the residual datasets and average whole-brain mask as inputs, and creates Monte-Carlo simulations of noise datasets which have the specified smoothness within the mask. It then creates a frequency distribution of noise-cluster sizes and advises the *P*-value and cluster extent required to control the FWER at a chosen level.

Analysis 3: Group-Level Conjunction Analysis for Force-Only and Pain-Only Trials

To identify the overlapping brain areas between the FO and PO conditions at the group level, all active voxels in the FWER-corrected

group-level statistical maps were set to 1 in the FO condition and to 2 in the PO condition. The resulting maps were added and voxels with a value of 3 were identified as overlapping voxels.

Analysis 4: Group-Level Voxel-Wise Contrasts Between Force-Pain and Other Conditions

The voxel-wise contrasts between FP and all other conditions were calculated by subtracting the regression-coefficient maps for FO, PO, and FW conditions from the regression-coefficient map for the FP condition. The resulting contrast maps (FP-FO, FP-PO, and FP-FW) were averaged across subjects, and *t*-statistics were calculated for each voxel to generate group-level statistical maps for each contrast. Each group-level contrast was thresholded to reject voxels with *P*-values >0.005 and clusters <567 mm³ in volume in order to keep the FWER <0.05.

Analysis 5: Individual-Level Conjunction Analysis for Force-Only and Pain-Only Trials

To identify the overlapping brain areas between the FO and PO conditions at the individual level, the procedure outlined in Analysis 3 was employed using each subject's FO and PO statistical maps, which had been corrected by rejecting voxels with *P*-values of >0.005 and clusters of <621 mm³ in volume in order to keep the FWER of <0.05. Owing to the variation in smoothness of the residual datasets across individuals, the extent threshold was found to vary from 540 to 621 mm³ across subjects. A conservative approach was taken and the most stringent threshold of 621 mm³ was used across all subjects. In addition to anatomical landmarks, the Harvard-Oxford Human Cortical Atlas (Desikan et al. 2006) was used to identify clusters of activity in the MCC, and the Human Motor Area Template (Mayka et al. 2006) was used to identify clusters of activity in supplementary motor area (SMA) and pre-SMA.

Previous evidence shows that the location of motor activity in the MCC is influenced by the presence or absence of a paracingulate sulcus. Whereas the location of the hand area of the middle cingulate region is similar in individuals with and without a paracingulate sulcus, other regions related to tongue and eye movements are shifted superior in individuals with a paracingulate sulcus (Amiez and Petrides 2014). To determine whether the variability of the sulcal anatomy in the cingulate region influenced the location of overlap in FO and PO conditions, we organized data from individual subjects according to whether a paracingulate sulcus was present (left hemisphere and right hemisphere) or absent (left hemisphere and right hemisphere). Given that our motor task and pain processing task primarily involved the hand, we did not expect the sulcal anatomy to influence the location of overlap between FO and PO conditions. Such a finding would support previous evidence by Amiez and Petrides (2014).

Results

Analysis 1: Force Data and Thermal Ratings

Figure 2*A* shows the mean force amplitude and Figure 2*B* shows the mean force error for the FO (blue bars), FW (green bars), and FP (black bars) conditions. Examination of the

force-traces from the PO condition showed that no force was produced during these trials, and so the PO condition was not included in the force analysis. Figure 2*A,B* shows nonsignificant effects of thermal stimulation on force amplitude (Fig. 2*A*: $F_{2,28} = 3.284$, $P = 0.052$) and force error (Fig. 2*B*: $F_{2,28} = 0.45$, $P = 0.641$). Although the effect of thermal stimulation on force amplitude approached significance, the mean grip-force differed by <0.07% of MVC between conditions. Nonsignificant differences were also evidenced for force variability ($F_{2,28} = 0.60$, $P = 0.553$), and the duration of force ($F_{2,28} = 1.62$, $P = 0.216$). Taken together, these findings show that force production was similar across FO, FW, and FP conditions.

Figure 2*C* shows the thermal stimulation ratings for the PO (red bars), FW (green bars), and FP (black bars) conditions. The temperature for warm stimulation was 40 ± 2.8 °C, and for pain-eliciting stimulation was 45 ± 1.64 °C. Subjects made ratings after every trial. The mean rating for the FO trials was zero for all subjects, and these data are not included in Figure 2*C* or the rating data analysis. Consistent with the pattern of data shown in Figure 2*C*, the corresponding 1-way ANOVA revealed significant differences between the conditions ($F_{1,26,17.67} = 173.22$, $P < 0.001$), with PO and FP trials rated as more painful than FW trials (P -values <0.001). Ratings for PO and FP trials were not significantly different ($t_{(14)} = -1.608$, $P = 0.130$), suggesting that the 2 tasks were matched in their subjective pain intensity.

Analysis 2: Brain Activation During Force-Only and Pain-Only Trials

The brain areas that were active at the group level during FO trials are presented in Table 1, and those during PO trials are presented in Table 2. Volume size, center-of-mass (CoM) coordinates, peak *t*-value, and the number of subjects who showed activity in each region are reported. Consistent with previous evidence, the brain areas recruited during the FO condition included left primary motor cortex (M1), dorsal premotor area (PMd), SMA-proper, aMCC, pMCC, insula, putamen, inferior parietal lobe, ventral-lateral thalamus, and lobules VI and VIIb in the cerebellum (Ebner and Fu 1997; Debaere et al. 2003; Krakauer et al. 2004; Lee and van Donkelaar 2006; Dannenberg et al. 2009; Roitman et al. 2009; Coombes et al. 2010, 2011; Coxon et al. 2010). Consistent with previous evidence, the brain areas recruited during the PO condition included aMCC, ACC, insula, pre-SMA, putamen, caudate, secondary somatosensory cortex, and lobules VI and VIIb in the cerebellum (Ploghaus et al. 2003; Wager et al. 2004; Apkarian et al. 2005;

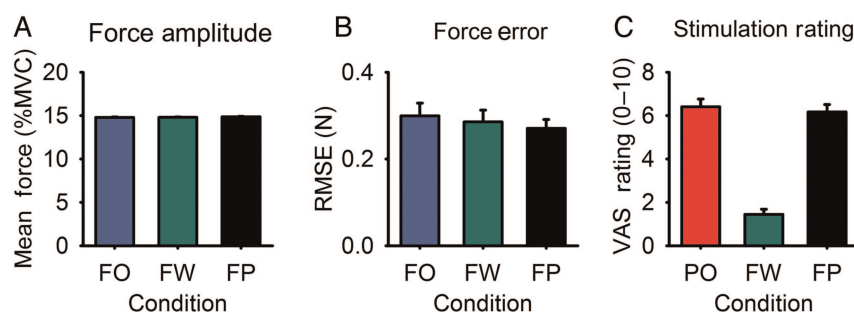


Figure 2. Mean force production and mean stimulation ratings. Conditions are color coded: FO (blue bars), PO (red bars), FW (green bars), and FP (black bars). (A) Mean force amplitude was not significantly different between conditions. (B) Force error, as indexed by root mean square error (RMSE), was not significantly different between conditions. (C) Mean stimulation ratings were significantly different between conditions. PO and FP ratings were each greater than FW ratings. FP and PO ratings were not significantly different from each other. Error bars represent standard error.

Table 1

The anatomical regions of localization, peak *t*-scores, MNI coordinates (CoM), and volume of each cluster identified at the group level for the FO condition

Region of activation	Peak <i>t</i> -score	CoM coordinates (MNI)			Volume (mm ³)	Subjects
		<i>x</i>	<i>y</i>	<i>z</i>		
Medial						
Cingulate cortex	7.89	5	13	45	648	13/15
SMA-proper	7.68	0	−3	60	2997	15/15
Pre-SMA	7.81	1	6	48	891	13/15
Left hemisphere						
Left insula	8.63	−43	2	6	4212	9/15
Left putamen	6.70	−27	−4	2	4887	10/15
Left thalamus (VLN)	8.00	−14	−19	9	1134	6/15
Left M1/PMd	6.32	−38	−15	56	4374	15/15
Left middle occipital gyrus	5.74	−28	−92	−3	2187	15/15
Left middle occipital gyrus	5.43	−42	−72	−1	1215	15/15
Left inferior parietal lobule	6.22	−30	−51	57	945	14/15
Left cerebellum, lobule VIIb	8.70	−18	−75	−47	1998	11/15
Right hemisphere						
Right insula	5.63	39	11	6	2106	12/15
Right putamen	6.75	28	3	5	783	6/15
Right thalamus (VLN)	6.21	12	−19	7	837	4/15
Right M1	4.87	38	−6	56	972	14/15
Right middle occipital gyrus	7.32	30	−96	−3	648	15/15
Right middle occipital gyrus	7.02	35	−85	4	2619	15/15
Right inferior iparietal lobule	5.54	45	−39	49	1350	13/15
Right inferior parietal lobule	5.34	33	−46	52	972	13/15
Right cerebellum, lobule VIIb	5.77	16	−74	−45	1782	11/15
Right cerebellum lobule VI	5.60	23	−56	−21	1107	11/15
Right inferior frontal gyrus	8.53	58	9	28	4131	15/15
Right inferior temporal gyrus	7.44	47	−62	−1	2970	15/15
Right postcentral gyrus	6.81	62	−22	38	2835	14/15
Right postcentral gyrus	4.76	63	−32	22	729	14/15

Note: The number of subjects who showed activity in each region is shown in the far right column. SMA, supplementary motor area; VLN, ventrolateral nucleus.

Table 2

The anatomical regions of localization, peak *t*-scores, MNI coordinates (CoM), and volume of each cluster identified at the group level for the PO condition

Region of activation	Peak <i>t</i> -score	CoM coordinates			Volume (mm ³)	Subjects
		<i>x</i>	<i>y</i>	<i>z</i>		
Medial						
Cingulate cortex	7.30	0	26	31	8694	14/15
SMA-proper	7.44	−1	0	61	1215	11/15
Pre-SMA	7.23	0	12	57	2187	11/15
Left hemisphere						
Left insula	7.00	−47	−13	13	16794	12/15
Left putamen	5.36	−26	−5	8	648	8/15
Left putamen	6.70	−22	8	−6	2619	8/15
Left caudate	4.61	−16	17	6	864	8/15
Left postcentral gyrus	6.75	−60	−33	24	1971	11/15
Left inferior parietal lobule	6.04	−39	−40	47	1890	12/15
Left inferior parietal lobule	5.20	−28	−52	51	783	12/15
Left cerebellum, lobule VIIb	6.61	−21	−74	−50	1944	9/15
Left middle occipital gyrus	6.40	−30	−91	−1	1512	14/15
Left middle occipital gyrus	6.20	−45	−70	3	783	12/15
Right hemisphere						
Right insula	6.80	39	16	3	3672	12/15
Right putamen	6.11	24	10	−5	1404	6/15
Right caudate	6.80	11	8	8	621	7/15
Right caudate	5.24	13	17	0	567	7/15
Right postcentral gyrus	8.48	62	−28	20	2241	12/15
Right inferior parietal lobule	6.22	50	−35	46	5022	14/15
Right cerebellum, lobule VI	5.31	33	−60	−23	621	7/15
Right middle occipital gyrus	7.01	31	−87	−2	2700	14/15
Right inferior occipital gyrus	4.62	43	−76	−10	675	11/15
Right superior occipital gyrus	6.25	28	−68	35	675	8/15
Right superior temporal gyrus	6.46	40	5	−13	1350	12/15
Right inferior frontal gyrus	7.00	55	5	31	2754	11/15

Note: The number of subjects who showed activity in each region is shown in the far right column. SMA: supplementary motor area.

Tracey and Mantyh 2007; Borsook et al. 2010; Moulton et al. 2011; Wager et al. 2013).

Analysis 3: Group-Level Conjunction Analysis for Force-Only and Pain-Only Trials

Our first goal was to determine whether overlapping functional activity would be evident in aMCC in the same group of subjects during separate trials of force production and pain processing. Hence, a conjunction analysis was conducted on the group activation maps for the FO and PO conditions. To identify the overlapping brain areas between the FO and PO conditions, all active voxels in the FWER-corrected group-level statistical maps were set to 1 in the FO condition and to 2 in the PO condition. The resulting maps were added and voxels with a value of 3 were identified as overlapping voxels. In Figure 3, blue voxels represent significantly active areas during the FO condition, red voxels represent significantly active areas during the PO condition, and green voxels represent the overlap. Significant clusters of overlap were found in the bilateral aMCC, pre-SMA, and SMA-proper. Additional clusters were identified in areas that included bilateral anterior insula (AI), left putamen, bilateral middle occipital gyrus, bilateral middle temporal gyrus, right precentral gyrus, right inferior parietal lobe, right postcentral gyrus, right superior temporal gyrus, and left lobule VIIb in the cerebellum. Volume and coordinates of the CoM of each cluster identified in the group-level conjunction analysis are presented in Table 3.

Analysis 4: Group-Level Voxel-Wise Contrasts Between Force-Pain and Other Conditions

Our second goal was to identify regions that showed changes in activity when force and pain processes were coupled in the same trial. Separate *t*-tests were used to compare activity during the FP condition with that during the FO, PO, and FW conditions. Labels, cluster size, and coordinates of brain regions identified in each contrast are presented in Table 4. Comparison of brain activity between FP and FO conditions revealed increased activity during the FP condition in the pain processing network in regions including ACC and left posterior insula. Activity in the ACC is commonly associated with pain processing and contributes to the neurologic signature of pain (Wager et al. 2013). Activity in the contralateral (left) posterior insula is also consistent with previous findings that identify this region as one of the few lateralized pain processing areas (Wager et al. 2013). In addition, as shown in Figure 4 (FP-FO), increased activity was found in the aMCC extending dorsally into the pre-SMA, consistent with the area of overlap identified in the group-level conjunction analysis. When compared with the PO condition, increased activity was found for the FP condition in the contralateral motor network, and other regions including left M1 and left ventral thalamus. Increased activity was also found in more posterior regions of aMCC extending dorsally into SMA-proper (Fig. 4: FP-PO). Increased activity during the FP condition when compared with the FW condition was found in the aMCC and lobule VI in the cerebellum (Fig. 4: FP-FW). For each contrast, percent signal change (PSC) was extracted from the identified clusters, and this is shown in the bar graphs. The FP condition shows a greater PSC when compared with the FO, PO, and FW conditions.

These findings are important for 2 reasons. First, the areas that showed increased activity in the FP condition in all 3

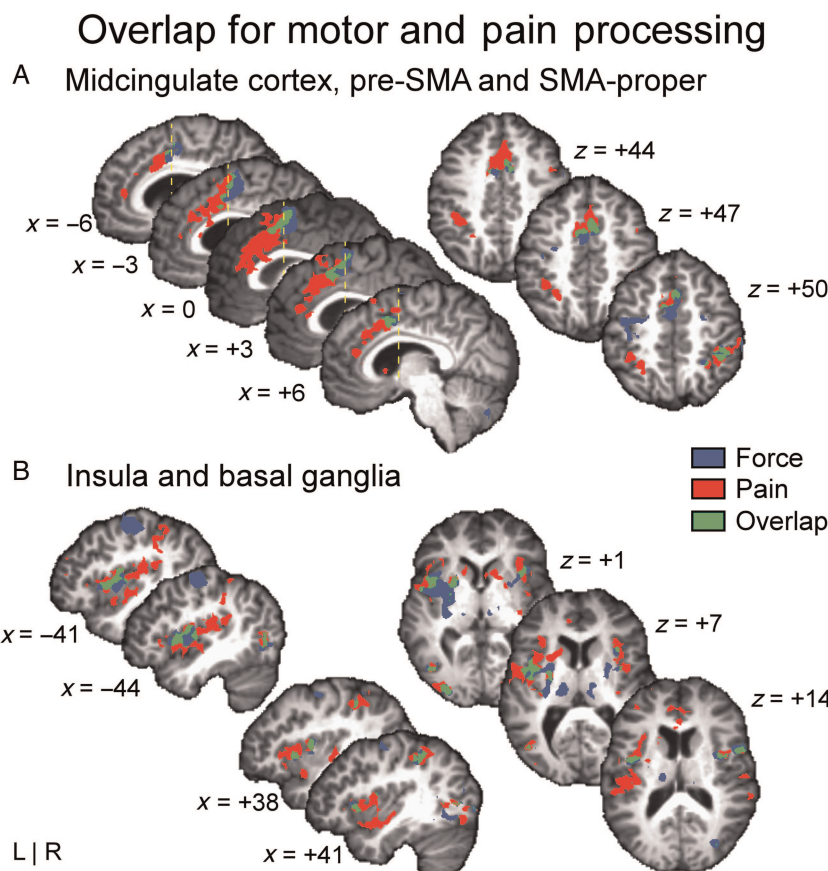


Figure 3. Group-level conjunction analysis. Significantly active regions at the group level in the FO condition are shown in blue, in the PO condition are shown in red, and the overlapping regions are shown in green. MNI coordinates of the sagittal and axial anatomical slices are indicated beside each slice. To identify the overlapping regions, all active voxels in the FWER-corrected statistical group maps were set to 1 in the FO condition (blue) and to 2 in the PO condition (red). The resulting maps were added and voxels with a value of 3 were identified as overlapping voxels (green). Conjunction analyses in MCC, pre-SMA, and SMA-proper (A), and insula and basal ganglia (B) are overlaid on sagittal and axial anatomical slices. The data are shown in the neurologic space. The yellow dotted line in (A) shows the plane of the anterior commissure. Coordinates for the CoM and volume of each overlapping cluster are presented in Table 3. The maps were resampled to a resolution of $1 \times 1 \times 1$ mm for presentation purposes only.

Table 3

Group-level conjunction analysis between FO and PO conditions

Region of activation	CoM (MNI)			Volume (mm ³)
	x	y	z	
Medial				
aMCC	0	9	48	945
SMA-proper	1	0	62	756
Pre-SMA	2	7	57	324
Left hemisphere				
Left insula	-44	1	8	2160
Left putamen (anterior)	-24	9	1	729
Left putamen (posterior)	-26	-5	5	432
Left middle temporal gyrus	-44	-71	4	351
Left middle occipital gyrus	-28	-92	-2	1080
Left cerebellum, lobule VIII	-15	-76	-45	621
Right hemisphere				
Right insula	38	18	4	432
Right insula	39	2	12	324
Right middle temporal gyrus	47	-63	5	729
Right middle occipital gyrus	34	-87	-2	1188
Right precentral gyrus	57	7	29	1620
Right postcentral gyrus	62	-21	37	756
Right superior temporal gyrus	63	-32	22	567
Right inferior parietal lobule	43	-41	49	1242

Note: The regions of activation, MNI coordinates (CoM), and volume of each cluster identified (FWER < 0.05) are listed.

aMCC, anterior midcingulate cortex; SMA, supplementary motor area.

Table 4

Group-level contrasts of FP versus FO, PO, and FW conditions

Contrast	Region	CoM (MNI)			Volume (mm ³)
		x	y	z	
FP-FO	aMCC	-1	13	39	1890
	Pre-SMA	-7	11	56	270
	Anterior cingulate cortex	-1	33	27	540
	Left posterior insula	-55	-20	14	864
	Left superior parietal lobule (SPL)	-19	-71	43	1566
	Left cerebellum lobule VI	-27	-58	-24	2538
	Left cerebellum lobule V	-5	-59	-7	1134
FP-PO	SMA-proper	-3	-5	57	783
	Left primary motor cortex (M1)	-44	-20	52	3132
	Left thalamus	-13	-19	7	918
	Left cerebellum lobule VIIIa	-15	-69	-48	1377
	Right primary motor cortex (M1)	39	-8	55	1458
	Right somatosensory cortex (S1)	62	-17	36	729
	Right inferior parietal lobule (IPL)	37	-43	54	2025
FP-FW	Right middle occipital gyrus (MOG)	37	-85	4	810
	aMCC	0	15	39	540
	Right cerebellum lobule VI	15	-80	-12	297

Note: The anatomical regions, MNI coordinates (CoM), and volume size of significant clusters (FWER < 0.05) for each contrast are listed.

aMCC, anterior midcingulate cortex; SMA: supplementary motor area.

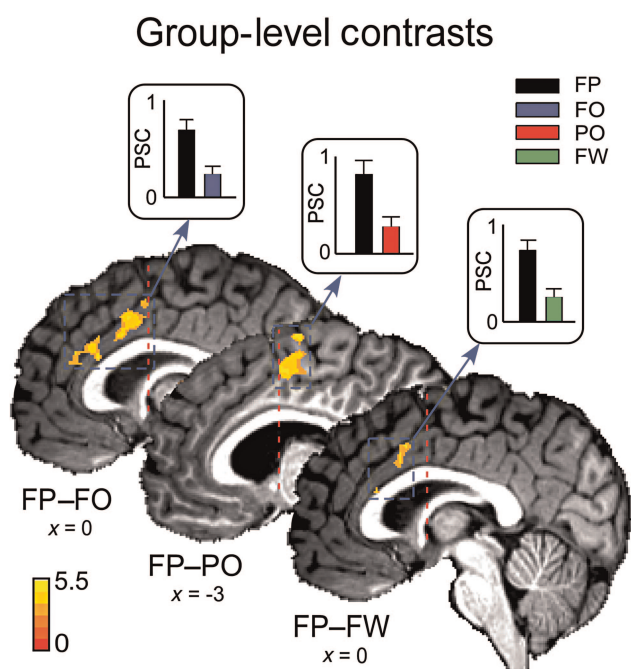


Figure 4. Group-level contrasts of FP versus FO, PO, and FW conditions. All 3 contrasts converged to show higher activity in the FP condition on the medial wall of the hemisphere. All regions identified in each of these contrasts are presented in Table 4. The color scale in the bottom-left represents t -values for the statistical maps that are overlaid on the anatomical images. PSC in the respective regions is shown in the bar graphs, where error bars represent standard error. The FP condition shows a greater PSC when compared with the FO, PO, and FW conditions, respectively. The red dotted line shows the plane of the anterior commissure. The maps were resampled to a resolution of $1 \times 1 \times 1$ mm for presentation purposes only.

contrasts were on the medial wall of the hemisphere. This finding is consistent with the overlapping activity identified in the group-level conjunction analysis and provides additional evidence for the importance of medial wall regions when motor and pain processing occur simultaneously. Secondly, none of the other regions that were identified in the conjunction analysis (left and right AI, left putamen, bilateral middle temporal gyrus, and middle occipital gyrus) were found to be different in the FP condition versus the other conditions. Hence, although insula and putamen are engaged when force and pain processing occur separately, they do not show increased activity when both processes occur simultaneously. Consistent visual feedback across conditions may account for the nonsignificant differences in the motion processing areas of extrastriate visual cortex. No regions were found to be less active in the FP condition as compared to all other conditions in any of the 3 contrasts.

Analysis 5: Individual-Level Conjunction Analysis for Force-Only and Pain-Only Trials

Our third goal was to determine whether overlapping functional activity would be evident in the aMCC in the same individual during separate trials of force production and pain processing. Group-level analyses converged to identify a region on the medial wall of the hemisphere that is positioned in the aMCC. However, one cannot assume from the group-level analyses that the same medial wall regions are engaged during motor and pain processing at the individual level. We therefore conducted conjunction analyses for each individual.

Individual-level conjunction analysis

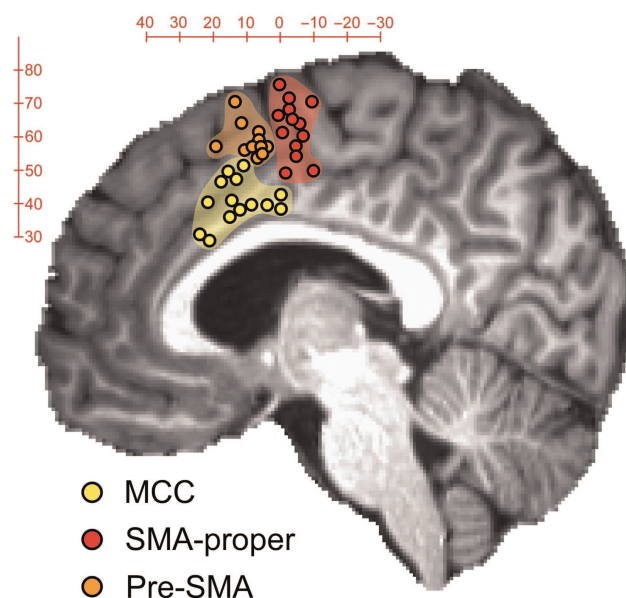


Figure 5. Individual-level conjunction analysis. Circles overlaid on the anatomical scan denote CoM of clusters that were significantly active during FO and PO conditions for each individual. In the MCC, 14 of the 15 subjects showed overlapping activity, and each of these subjects is represented by a yellow circle. Red circles represent the CoM of overlap clusters for the 13 subjects who showed overlapping activity in SMA-proper, and orange circles represent the CoM of overlap clusters of the 11 subjects in pre-SMA. Since the CoMs for different subjects were spread across different sagittal planes, all coordinates were projected onto the plane $x = 3$ mm for easier visualization. The x -coordinates of CoM of all clusters ranged from $+9$ to -9 mm. The exact coordinates and volume size for overlapping clusters for each individual are given in Table 5. MCC: midcingulate cortex; SMA: supplementary motor area.

All active voxels in an individual's FWER-corrected statistical map were set to 1 in the FO condition and to 2 in the PO condition. The resulting maps were added, and voxels with a value of 3 were identified as overlapping voxels. Each circle in Figure 5 represents the CoM of the overlap cluster between the FO and PO conditions for one subject. Yellow circles represent overlap in the MCC, red circles represent overlap in SMA-proper, and orange circles represent overlap in pre-SMA. In MCC, 14 of the 15 (93%) subjects showed overlap. In SMA-proper, 13 subjects showed overlap (87%), and 11 showed overlap in pre-SMA (73%). CoM coordinates of the overlapping cluster and the volume of the cluster for each subject are listed in Table 5.

We next examined the patterns of overlapping functional activity in the MCC when taking into account individual differences in the sulcal/gyral morphology of the cingulate cortex. The cingulate sulcus is divided into several antero-posterior segments. In the medial frontal lobe, an additional cingulate sulcus is present in a subset of individuals and is referred to as the paracingulate sulcus, which is superior to and shorter than the cingulate sulcus. Three vertical sulci can typically be identified with each one joining either the cingulate sulcus or the paracingulate sulcus. Most posterior is the paracentral sulcus, followed by the preparacentral sulcus, and then the most anterior vertical paracentral sulcus (Vogt et al. 1995; Paus, Tomaiuolo, et al. 1996; Fornito et al. 2008).

Table 5
Individual-level conjunction analysis

Subject	MCC				SMA-proper				Pre-SMA			
	CoM (MNI)			Volume (mm ³)	CoM (MNI)			Volume (mm ³)	CoM (MNI)			Volume (mm ³)
	x	y	z		x	y	z		x	y	z	
1	−9	5	39	108	−3	−5	63	54	−	−	−	−
2	2	0	38	513	3	−3	64	2025	9	7	60	459
3	5	14	38	108	0	−4	56	1674	9	7	53	594
4	−1	14	37	135	2	−2	67	864	1	6	56	567
5	3	18	46	189	4	−2	69	1458	5	8	53	432
6	−2	17	50	216	0	0	50	54	7	11	55	189
7	9	14	48	297	2	1	65	297	2	12	63	378
8	−1	22	40	540	2	1	74	270	3	14	69	1404
9	−3	11	39	108	1	0	60	405	−	−	−	−
10	−3	15	36	108	−	−	−	−	−	−	−	−
11	1	11	51	270	−2	−4	54	1566	−3	7	54	1134
12	0	15	41	1890	−1	−6	59	4050	1	7	56	1134
13	3	0	42	81	1	−2	49	162	2	7	58	108
14	3	22	29	594	−5	−9	49	378	2	20	56	972
15	−	−	−	−	−	−	−	−	−	−	−	−
Mean	1	13	41	368	0	−3	60	1020	3	10	58	670
SD	4	7	6	472	2	3	8	1141	4	4	5	426

Note: Overlap clusters identified in MCC, SMA-proper, and pre-SMA for each subject are listed. MNI coordinates (CoM) and volume sizes of the largest cluster in each area are reported for each subject. The conjunction analysis identified significantly active clusters (FWER < 0.05) that were common to FO and PO conditions at the individual-subject level. The mean and standard deviation of the coordinates and cluster volume for each area are also reported. “−” denotes that a subject did not have overlap in an area. MCC, midcingulate cortex; SMA, supplementary motor area.

The most prominent difference between the cingulate cortices of individuals was the presence or absence of a paracingulate sulcus. The presence of a paracingulate sulcus was based on the identification of a horizontal sulcus running dorsal and parallel to the cingulate sulcus for at least 25 mm and was observable for at least 3 contiguous sagittal slices. The absence of a paracingulate sulcus was confirmed if there was no indication of a paracingulate sulcus in the relevant slices (Buda et al. 2011; Amiez and Petrides 2014). All of our subjects showed a cingulate sulcus bilaterally and 12 of them (subjects 1–5, 7–8, 11–15) showed a paracingulate sulcus in at least one hemisphere (subjects 3, 4, and 12 displayed it bilaterally). Thus, a paracingulate sulcus was observed in 80% of our subjects, which is consistent with the previous literature (Paus, Otaky, et al. 1996; Paus, Tomaiuolo, et al. 1996; Fornito et al. 2008). To examine whether a paracingulate sulcus influenced our findings, we used T_1 -weighted anatomical images to classify individuals into 4 separate groups according to the presence or absence of the paracingulate sulcus in their left or right hemisphere. Figure 6 shows data for the individual-level conjunctions in the left and right hemispheres for subjects with and without a paracingulate sulcus. Table 6 lists the coordinates and volume size of the conjunction cluster for each individual. The average x -, y -, and z -coordinates for the FO–PO overlap in the left hemisphere for individuals with a paracingulate sulcus are −4, 13, 40, which are similar to the average coordinates for individuals without a paracingulate sulcus: −3, 9, 42. Similarly, in the right hemisphere, the average coordinates for individuals with a paracingulate sulcus are 3, 9, 41, which are similar to the average coordinates for individuals without a paracingulate sulcus: 3, 12, 42. These observations suggest that the presence of a paracingulate sulcus did not influence the location of the overlap in functional activity between the FO and PO conditions.

Discussion

Our experiment produced 3 novel results. First, we provide group-level evidence of overlapping functional activity during motor control and pain processing in several areas, including aMCC, pre-SMA, and SMA-proper. Secondly, we present group-level evidence of increased activity in aMCC, pre-SMA, and SMA-proper when motor and pain processes occurred simultaneously. Thirdly, at the individual-subject level, we demonstrate overlapping activity in the aMCC in 93% of our subjects, in SMA-proper in 87% of our subjects, and in pre-SMA in 73% of our subjects. Consistent with predictions based on the somatotopy of hand function in the cingulate cortex, we also show that the areas of overlap in the aMCC at the individual level were not influenced by the presence or absence of a paracingulate sulcus. These observations extend prior anatomic, physiologic, and imaging data by providing direct evidence in humans that motor control and pain processing converge in the same regions of the medial wall of the hemisphere.

We show that the FO condition engaged brain regions consistent with the visuomotor network (Prodoehl et al. 2009; Coombes et al. 2010, 2011), and the PO condition engaged brain regions consistent with the pain processing network (Ploghaus et al. 2003; Apkarian et al. 2005; Vogt 2005; Tracey and Mantyh 2007; Wager et al. 2013). We first consider our finding that the same region of aMCC was active during separate motor control and pain trials. This is consistent with the meta-analysis reported by Dum et al. (2009), who showed overlapping functional activity in aMCC based on the amalgamation of peak voxel locations identified in neuroimaging studies that separately manipulated motor and pain tasks in different groups of subjects. The meta-analysis revealed functional overlap across much of the cingulate cortex, with a noticeable concentration of overlapping voxels positioned at the aMCC–pMCC border. Our observations are in line with this meta-analysis and identify aMCC within the cingulate cortex as a sensorimotor region where motor control and pain processing converge. The location of activity within aMCC identified

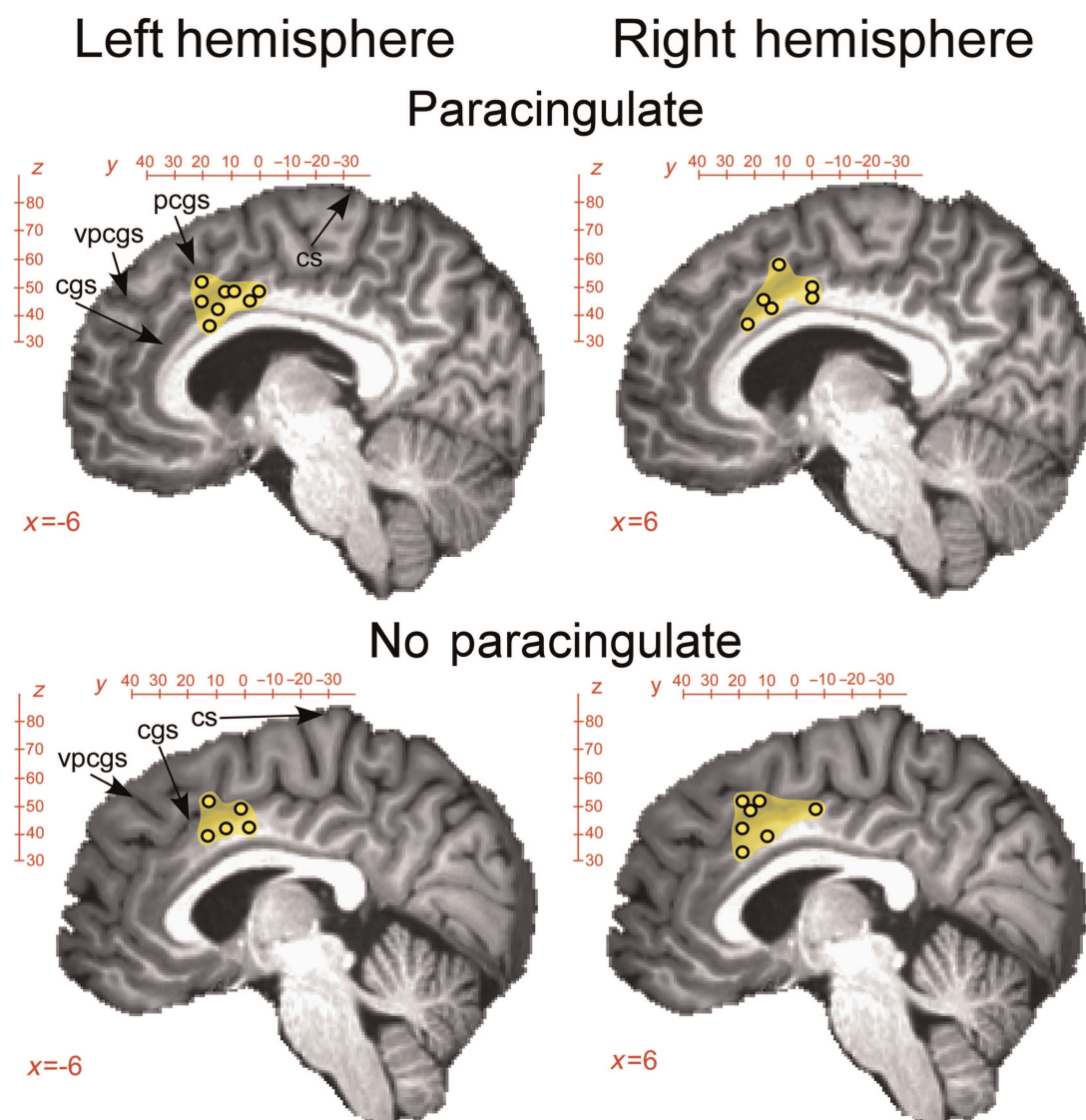


Figure 6. Conjunction maps grouped according to cingulate morphology. The individual-level conjunction maps were split into the left and right hemispheres. The resulting maps were then subdivided based on whether the morphology of the hemisphere showed a paracingulate sulcus or not. Circles overlaid on the anatomical scans denote the CoM of the overlap clusters that were significantly active during FO and PO conditions for each individual. Since the CoM for different subjects were spread across different sagittal planes, all CoMs were projected onto sagittal slices at $x = 6$ mm (right) and $x = -6$ mm (left) for easier visualization. The exact coordinates and volume size for overlapping clusters for each individual are given in Table 6. The sulci of interest are labeled as: cs, central sulcus; cgs, cingulate sulcus; pcgs, paracingulate sulcus; vpcgs, vertical paracingulate sulcus.

in the current study is consistent with the arm representation area of ventral cingulate motor area in monkey (Luppino et al. 1991) and with the provisional hand and arm region in posterior RCZ in humans (Picard and Strick 1996; Nee et al. 2011). A recent human neuroimaging study used a range of motor tasks to identify 3 distinct somatotopically organized regions in the MCC (Amiez and Petrides 2014). The cluster of overlapping activity in aMCC in the current study was similar in the x -direction and z -direction to the middle cingulate region identified by Amiez and Petrides, although our cluster was approximately 9 mm anterior. Importantly, we found no evidence of overlap in the posterior and anterior regions of cingulate cortex that were identified by Amiez and Petrides, which suggests that, in the context of hand-related motor control and pain processing, sensorimotor integration is limited to aMCC.

The conjunction analysis and the voxel-wise contrasts together suggest that, in addition to motor and pain-related

processes, aMCC is also sensitive to changes in the intensity of a thermal stimulus. The cluster of activity identified in the FP-FW contrast, which controls for areas that are engaged during nonpainful thermal processing, is much smaller than the cluster identified in the FP-FO contrast and the overlapping activity identified in the conjunction analysis. Hence, while much of aMCC is engaged during sensory processing of a thermal stimulus, only a subregion of aMCC shows increased activity when the thermal stimulus is painful. This is consistent with previous evidence which associated dorsal and posterior regions of aMCC ($-3, 3, 51$) with stimulus perception, aMCC with cognitive processing and stimulus intensity ($-3, 21, 45$), and ventral aMCC areas that traverse the aMCC/pMCC border with pain intensity ($0, 18, 36$) (Buchel et al. 2002). Coordinates of the cluster of activity identified in the FP-FW contrast in the current study ($0, 15, 39$) span the cognitive processing and pain intensity regions previously identified. Other evidence

Table 6

Individual-level conjunction analysis for overlapping activity in the MCC grouped according to the presence or absence of a paracingulate sulcus in the left and right hemispheres

Subject	Paracingulate				No paracingulate			
	Left		Right		Left		Right	
	CoM (MNI)		CoM (MNI)		CoM (MNI)		CoM (MNI)	
	x	y	z	Volume (mm ³)	x	y	z	Volume (mm ³)
1	−9	5	39	108				
2								
3	0	18	30	27				
4	−1	14	37	135				
5	−6	9	44	54				
6								
7	−9	0	42	27				
8	−1	22	40	486				
9								
10								
11								
12	−3	12	41	1107				
13								
14	−2	22	46	297				
15	—	—	—	—				
Mean	−4	13	40	280				
SD	4	8	5	370				

Note: The conjunction analysis identified significantly active clusters (FWER < 0.05) in the FO and PO conditions. MNI coordinates of the CoM of each cluster and cluster volume are reported. The mean and standard deviation of the coordinates and volume size for each cluster for each group are presented. Each hemisphere of a subject is represented in one column. “—” denotes the absence of overlap in a hemisphere. Subject 15 did not show overlap in any hemisphere.

from microelectrode single neuron recordings in humans also show that aMCC is engaged during attention demanding cognitive tasks, such as mental arithmetic and word generation (Davis et al. 2000), and functional imaging evidence has shown that the brain networks, including MCC, that support pain perception and cognition can be simultaneously active (Seminowicz and Davis 2007). An alternative dual-task explanation for our findings is that aMCC was more active during the FP condition because of an increase in attentional load when force was produced while experiencing and evaluating a thermal stimulus. Although the FW condition does control for the dual-task issue, it is clear from our data and other evidence that while much of aMCC may be involved in sensory processing, as shown by the larger overlap in the conjunction analysis, a much smaller subregion of this area scales in activity when force production and pain processing are coupled. Although our study was not designed to delineate cognitive or attentional processes from stimulus intensity processes, our findings do offer additional evidence for the multifaceted role that aMCC plays in the processing of sensory information (Shackman et al. 2011).

Although our findings converge with and extend recent reports of sensorimotor integration in aMCC (Dum et al. 2009; Pereira et al. 2010; Nee et al. 2011; Shackman et al. 2011; Amiez and Petrides 2014), they are notably different from the one other study that examined motor control and pain processing in the same human subject. Kwan et al. (2000) found evidence for a functional segregation in the aMCC, with ventral portions associated with pain processing and dorsal portions associated with motor processing. Our findings contrast these results because they show that the same region of aMCC is engaged during motor control and pain processing at the group level, as well as increased activity in this same region when force and pain processes occurred simultaneously as compared with separately. Hence, our data support an integrated view of aMCC function, which contrasts the segregated

view put forward by Kwan et al. (2000). This may be due to methodological differences. Kwan et al. used a 1.5-T scanner, collected functional data from 4 sagittal slices close to the midline, used a self-paced finger–thumb tapping task with no defined target force or means to verify task performance, and used a sinusoidal temperature profile to elicit pain. We used a 3-T scanner with whole-brain coverage, a visually guided precise grip-force task, and a constant temperature to elicit pain. We also temporally coupled force production and thermal stimulation in the same trial on the same hand, and evidence derived from this novel approach was consistent with our expectations based on data from separate motor control and pain processing conditions. The current findings are important because they provide new data that support an integrated rather than a segregated view of aMCC function (Shackman et al. 2011), and do so for the first time in the context of motor control and pain processing in humans.

Other evidence for an integrative role of aMCC comes from a meta-analysis of human neuroimaging work that identified functional activation common to negative affect, pain, and cognitive control in the aMCC (Talairach coordinates = 0, 12, 42; MNI coordinates = −1, 6, 43) (Shackman et al. 2011). Our MNI coordinates of 0, 9, 48 (Table 3) are strikingly close to those reported by Shackman et al., suggesting that our finding dovetails nicely with previous studies that have identified multimodal processing in the aMCC. Our findings are also consistent with animal and human studies that have identified anatomical and functional connections between aMCC and areas that include the primary motor cortex (Picard and Strick 1996), insula (Mesulam and Mufson 1982; Cauda et al. 2011), dorsolateral prefrontal cortex (Hatanaka et al. 2003), and the lateral basal nucleus of the amygdala (Morecraft et al. 2007), which together represent key regions of the motor, pain, cognitive, and emotional networks. Hence, aMCC is uniquely positioned to synthesize information from these networks to guide behavior. An understanding of how these networks are engaged and

integrated is essential if we are to better understand the supraspinal mechanisms that underlie how pain and movement influence each other in rehabilitation and performance contexts (Bank et al. 2013; Koltyn et al. 2013). Behavioral and neurophysiologic evidence have shown that pain influences decisions about the movements that we make and how we control those movements (Zedka et al. 1999; Falla et al. 2007; Thomas and France 2008; Farina et al. 2012). Taken together with previous evidence, our data suggest that functional activity in the aMCC relates to how motor function and pain processes are integrated (Peyron et al. 2007).

Individual-level analysis substantiated and extended our group-level findings by identifying that the same region of aMCC was active across pain and motor tasks in 14 of the 15 subjects. Coordinates for the CoM of 12 subjects were located in the aMCC, and coordinates of 2 subjects were located at the border of aMCC and pMCC ($y = 0$). We also examined whether individual differences in sulcal/gyral morphology influenced the location of overlap in aMCC. Our findings were consistent across all subjects independent of whether or not an individual had a paracingulate sulcus, with only small deviations in the average coordinates of overlapping motor–pain clusters between groups (Fig. 6 and Table 6). This is consistent with the findings of Amiez and Petrides (2014), who demonstrated that the hand area of the middle cingulate region was similar in individuals with and without a paracingulate sulcus, whereas other regions related to tongue and eye movements were shifted superior in individuals with a paracingulate sulcus.

In addition to the MCC, SMA-proper and pre-SMA were identified in our group-level and individual-level analyses. SMA-proper has dense connections with M1 and MCC and projects directly to the spinal cord, whereas pre-SMA does not have substantial connections with M1 and does not project to the spinal cord (Dum and Strick 1991, 1996). Early work detailing the function of these regions suggested that pre-SMA is involved in controlling internally guided movements, whereas SMA-proper is involved in controlling externally guided movements (Mushiake et al. 1991; Deiber et al. 1999). However, more recent findings show that both pre-SMA and SMA-proper are engaged during internally and visually guided movements (Picard and Strick 2003; Vaillancourt et al. 2003), with differences emerging as a function of motor planning. Although the SMA is traditionally thought of as a motor area, it is also engaged during pain processing experiments that do not explicitly manipulate motor function (Wager et al. 2013). Our group-level analysis of the PO condition is consistent with this observation (Table 2). Engagement of SMA-proper and pre-SMA during pain processing is often interpreted as engagement of the motor system in planning or producing a behavioral response to pain. Our findings are consistent with this interpretation and show that the same regions of SMA-proper and pre-SMA are engaged when pain and motor control processes occur separately, and this is in line with previous findings (Kwan et al. 2000). Although the role of SMA and its link to motor function in the context of pain is not well understood, we have previously shown that pre-SMA is recruited when force control is maintained in emotional contexts (Coombes et al. 2012), consistent with previous evidence linking emotion and motor function (Coombes et al. 2007a, 2007b, 2008, 2009; Schmidt et al. 2009). Hence, one interpretation of the current findings is that pre-SMA plays a role in controlling force

production in painful as well as emotional contexts. Although force output often remains constant even when an individual is feeling pain, the muscle coordination that drives force output is adapted (Hodges and Tucker 2011), and our observations suggest that pre-SMA and SMA-proper may be involved in this adaptation given their roles in the planning and execution of voluntary movement.

AI and putamen were also identified in the group-level conjunction analyses as processing both motor control and pain signals. Consistent with previous somatotopic maps of insula during limb-specific motor function and pain processing (Mutschler et al. 2009; Kurth et al. 2010), we found overlapping activity at the group level in the posterior region of AI, a finding that fits well with the shared resource model of insula as well as the awareness model of AI (Craig 2009). Although a role for the basal ganglia in motor processing is well established (Prodoehl et al. 2009), the association between basal ganglia function and pain has only recently gained traction (Borsook et al. 2010), and is supported by studies that link reductions in pain perception with putamen lesions and with electrical stimulation of the caudate and globus pallidus (Lineberry and Vierck 1975; Favre et al. 2000; Starr et al. 2011). Other evidence shows that acute pain activates the putamen bilaterally, and that a somatotopic organization for hand- and foot-related responses is present in the contralateral putamen, but not other basal ganglia nuclei (Bingel et al. 2004). Our group-level findings offer new evidence that identifies AI and putamen as common substrates for motor and pain processing. However, these areas are distinguished from medial wall areas because activity in these regions did not differ when force and pain were temporally coupled as compared to when they were not. Further investigation is necessary to explore the role that insula and basal ganglia play in the motor–pain relationship.

Pain changes the way that we move, and the way that we move can change how we experience pain (Hodges and Tucker 2011; Naugle et al. 2012; Bank et al. 2013; Paris et al. 2013). Behavioral and neurophysiologic evidence have associated pain with adaptations in muscle activity and with increased low-frequency variability in synaptic input to motor units during voluntary movement (Zedka et al. 1999; Falla et al. 2007; Farina et al. 2012). Other evidence has shown that perceived pain is reduced during or following isometric force production and during motor cortex stimulation (Leo and Latif 2007; Lima and Fregni 2008; Umeda et al. 2010; Reidler et al. 2012; Paris et al. 2013). However, the neural basis underlying the motor–pain relationship is not well understood (Borsook 2007; Bank et al. 2013). Our observations provide the first direct evidence that the same regions of aMCC and SMA in the medial wall of the hemisphere are components of the brain networks that underlie both motor control and pain processing. These findings provide a foundation for future studies to examine the neural basis of sensorimotor integration in the context of motor control and pain processing.

Notes

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